Fluid Flow & Piping Systems

Reynolds Number is a dimensionless quantity that describes the flow of a fluid in a pipe. It is defined as

$$Re = \frac{\rho v d}{\mu} = \frac{vd}{\nu}$$

where v is the velocity of the fluid, d is the diameter of the pipe and μ is the dynamic viscosity of the fluid. Reynolds number may also be written in terms of the kinematic viscosity, ν.

Reynolds number is typically in the thousands, and describes one of three different kinds of flow.

Laminar Flow $(Re < 2000)$

> Laminar flow refers to fluid flowing in layers, which may have different velocity magnitudes, but all have the same direction, and the layers do not disturb one another. Viscous forces are more present

Turbulent Flow $(Re > 4000)$

> Neighboring velocities are now no longer parallel, causing them to interfere and mix with one another. Dynamic forces are now more present.

Transition Flow $(2000 \leq Re \leq 4000)$

> As Re changes from 2000 (laminar) to 4000 (turbulent), flow gradually changes from these two flow types, however its uncommon to do calculations with these. Re = 2000 and Re = 4000 is also called the critical values of flow.

In a pipe, there is also some friction loss as fluid flows through. It is determined by the Darcy-Weisbach Equation, given as

$$H_f = \frac{4fL}{d}\left(\frac{v^2}{2g}\right)$$

where H_f is the frictional head loss in m, f is the friction factor (dimensionless), dependent on the fluid, and L is the length of the pipe.

v is also the average velocity of the fluid, given as

$$v = \frac{Q \text{ [average flow rate]}}{A}$$

The Darcy formula is not exact and is only a simplification; frictional forces are not fully proportional to v^2, and the friction factor is not fully a constant, but depends on Re.

There is also some pressure drop as fluid flows in a pipe, governed by the Hagen-Poiseuille equation,

$$\Delta P = Q \left(\frac{128 \mu L}{\pi d^4} \right)$$

but since $Q = vA = v \left(\frac{\pi}{4} d^2 \right)$, the drop in pressure can be rewritten as

$$\Delta P = \frac{32 \mu L v}{d^2}$$

Since $H_f = \frac{\Delta P}{\rho g}$, the Darcy equation can be rewritten as

$$H_f = \frac{32 \mu L v}{\rho g d^2}$$

which can be written in terms of Reynolds number, by

$$\begin{aligned} H_f &= \frac{32 \mu L v^2}{\rho v g d^2} \\ &= \frac{32 L v^2}{g d} \times \frac{\mu}{\rho v d} \\ &= \frac{32 L v^2}{g d} \times \frac{1}{Re} \end{aligned}$$

and can be rewritten closer to the original Darcy equation,

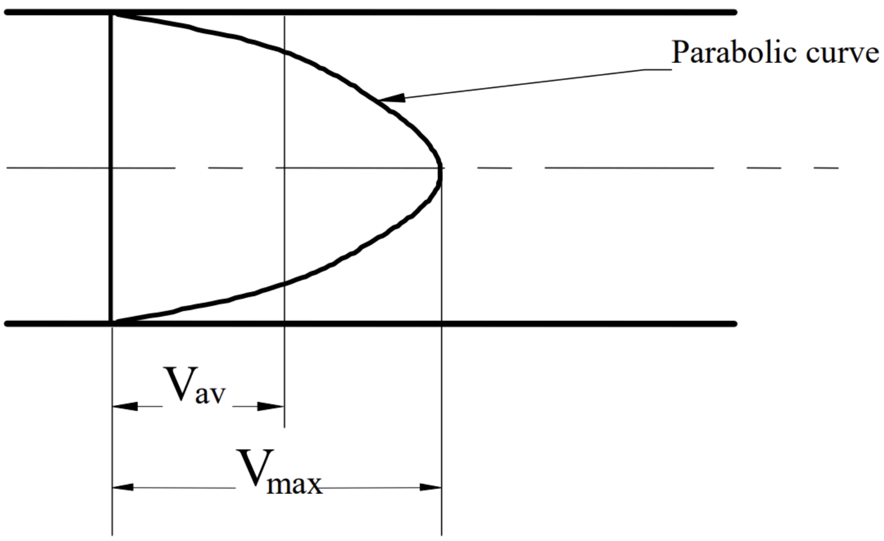
$$\begin{aligned} &= \frac{64 L v^2}{2 g d} \times \frac{1}{Re} \\ &= \frac{64 L}{d} \times \frac{v^2}{2g} \times \frac{1}{Re} \\ &= \frac{\frac{64}{Re} L}{d} \times \frac{v^2}{2g} \\ &= \frac{4 \left(\frac{16}{Re} \right) L}{d} \times \frac{v^2}{2g} \end{aligned}$$

and by comparision, we see that $f = \frac{16}{Re}$ (for laminar flow).

As mentioned earlier, a fluid flowing in a pipe will have different velocities at different layers. The distribution of these velocities is described by the velocity profile, a curve representing the velocities across the diameter of the pipe.

In laminar flow, the velocity profile is a parabola, where the average velocity is half of the maximum fluid velocity, or

$$V_{avg} = \frac{1}{2}V_{max}$$



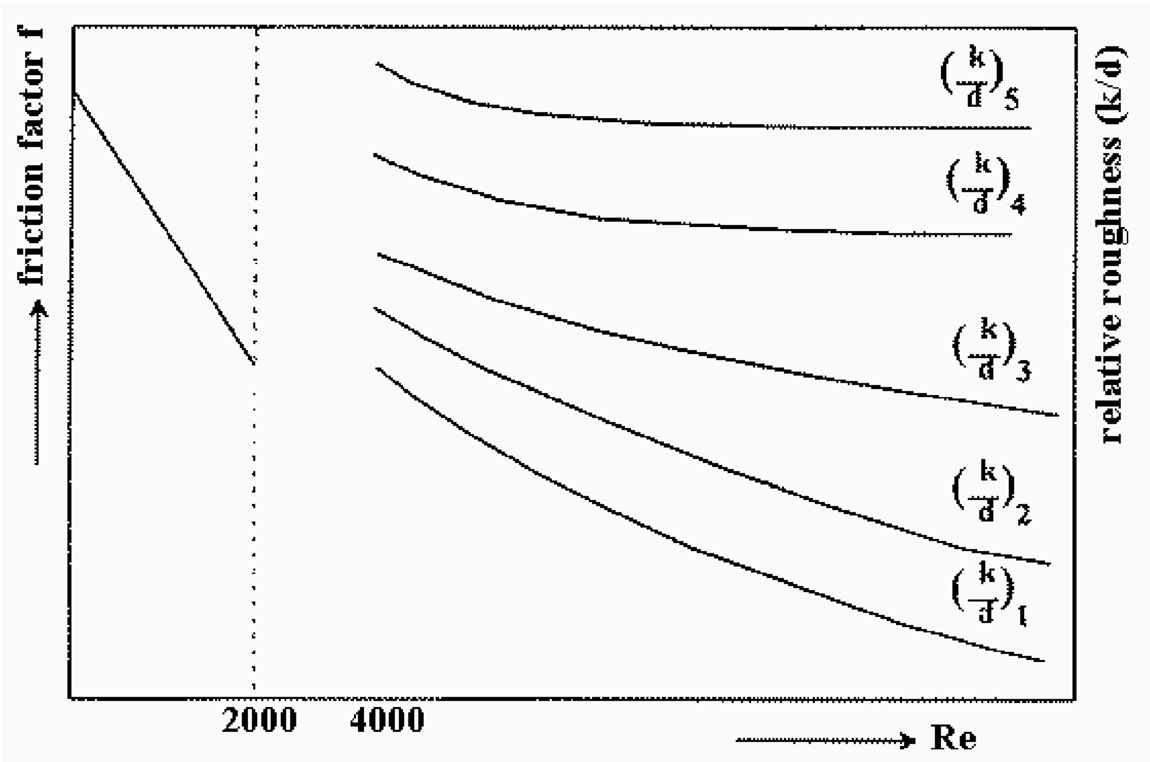
Turbulent flow's velocity profile is typically truncated, the front portion of the profile looks cut off, with a higher Reynolds number having a higher peak velocity (as $Re \propto v$), but the shape of the other velocities depend on the Reynolds number.

One reason why there are losses in pressure in a pipe is because of the roughness of the pipe. If we let the average height of the internal surface of the pipe's roughness projection be k, also called the absolute roughness, the relative roughness is given as

$$\text{Relative Roughness} = \frac{k}{d}$$

In laminar flow, the pipe roughness has no effect on frictional losses. However, turbulent flow has an increase in frictional losses due to higher friction, and the friction also affects Re.

Using relative roughness, the friction factor can be found, by the Moody Diagram.



The Moody diagram takes in the Reynolds number and the relative roughness, to give a curve and the value of the friction factor. These values are experimentally found, but for smooth pipes, of $4000 < Re < 10^5$, friction factor can be found by Blasius' formula,

$$f = 0.079Re^{-0.25}$$

which is only a fit of the friction factor, and not fully exact.

Other than frictional losses due to rough pipe surfaces, frictional losses can occur across valves and fittings. Hence, the head loss across a fitting is given as

$$h_f = \alpha \left(\frac{v^2}{2g} \right)$$

where v is the average velocity flow in the pipe, not at the fitting itself. The coefficient α is the frictional coefficient, dependent on the type of the fitting.

Valves & Fittings	α
90° bend	0.20
45° bend	0.10
Gate valve (fully open)	0.20
Globe valve (fully open)	8.00
Abrupt entry	0.50
Gradual entry	0.05
abrupt exit	1.00
Gradual exit	0.12
Inlet strainer and foot valve	2.80
Check valve	2.50

These values are for reference only, and depend on the actual design of the fitting itself.

If we equate this to the Darcy equation, we can find an equivalent of pipe that produces the same head loss. In other words,

$$h_f = \alpha \left(\frac{v^2}{2g}\right) = \frac{4fL_{eq}}{d}\left(\frac{v^2}{2g}\right)$$

The equivalent length is typically found through tables or through a nomogram. We can also solve for the equivalent length to get

$$\frac{4fL_{eq}}{d} = \alpha$$
$$L_{eq} = \frac{\alpha d}{4f}$$

So far, we have only discussed fluid flow in a single pipe. However, most systems may have multiple pipes arranged in series or parallel.

In series pipes, frictional losses may come up due to different diameters and roughness of the pipes. We can form two equations relating the volumetric flow and frictional loss of two series pipes, as

$$Q = Q_A = Q_B$$
$$H = H_A + H_B$$

Because there are only two equations, only two unknowns, volumetric flow or velocity, may be found.

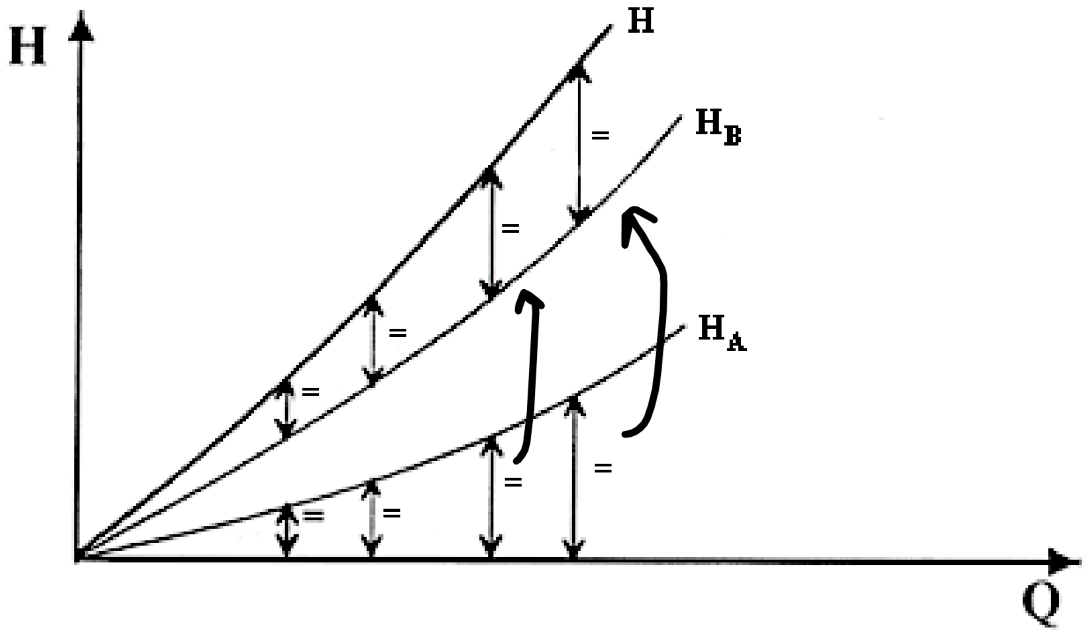
However, solving the equations may be complex, and is usually solved graphically instead.

Firstly, since the Darcy equation shows that head loss is proportional to v^2, and $Q = vA$,

$$H \propto Q^2 \Rightarrow H = CQ^2$$

Thus, $H_A = C_A Q_A^2$ and $H_B = C_B Q_B^2$.

If we plot H against Q, we see two parabolic curves for each pipe:

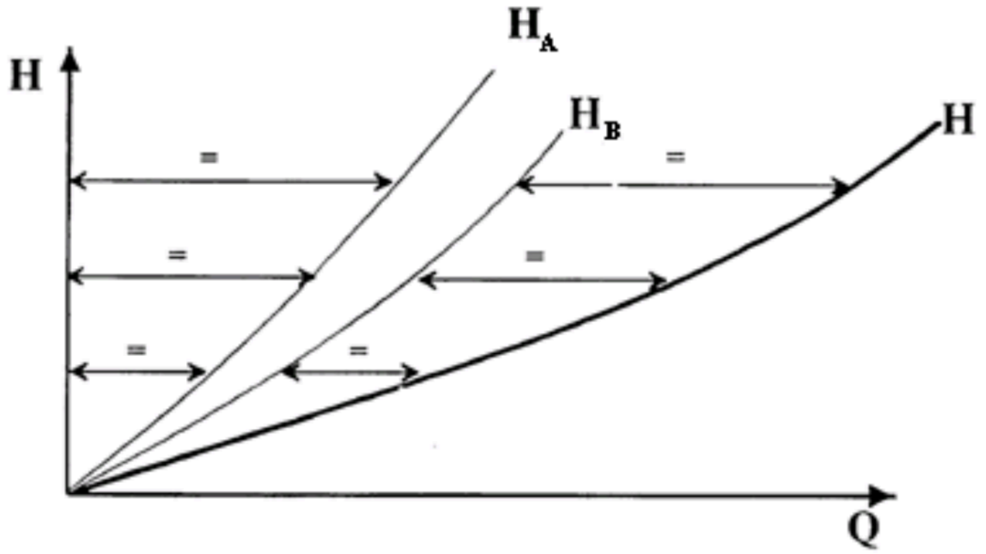


We know that $Q_A = Q_B$. Thus, if we know a flow rate, we can find the head loss for each pipe. To sum them up, we combine the vertical lengths by transferring them like so, to find the total head loss.

In parallel pipes, when two pipes split and join again, the energy at the joint must be equal for both pipes. This means that the frictional head loss across the parallel section of the pipe must be equal. Thus,

$$Q = Q_A + Q_B$$
$$H = H_A = H_B$$

Again, since $H = CQ^2$, we can solve for Q graphically instead, by joining the horizontal distances on the H-Q graph.



**For pumps, which have to overcome the frictional losses to pump the fluid through a pipe, the formula for the power is given as

$$\text{Power} = \frac{\rho g H_f Q}{\eta}$$